PHYSICS 1

UNIT 3: MOMENTUM, COLLISIONS & ROTATIONAL MOTION

Fall 2026

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Name: ______________________________

How to Use This Packet

Each section below has two parts, in the Cornell note-taking format, plus a summary box.

Notes column (right, wide)

Fill this in during lecture. Definitions, formulas, and key terms are partly blanked — write the missing words, symbols, or numbers as your instructor presents them.

Cue column (left, narrow)

These are the key questions and terms for the section. Use them right after class (or before a quiz) to test yourself: cover the notes column and see if you can answer each cue from memory.

Summary box

At the end of each section, write 2–3 sentences in your own words capturing the main idea. If you can't summarize it, you don't know it yet — go back and review.

Chapter 8: Linear Momentum and Collisions

8.1 Linear Momentum and Force

CUE / KEY QUESTIONS *Define linear momentum — is it a vector or scalar?* *SI unit of momentum?* *State Newton's Second Law in terms of momentum.*	Linear momentum (p) is the product of an object's mass and velocity: p = _________. It is a ___________ quantity, pointing in the same direction as the ___________. SI unit of momentum: __________________ (no separate named unit). Newton's Second Law, more generally: Fnet = ________________ — the net external force equals the rate of change of momentum. This reduces to F = ma when mass is constant.

SUMMARY — in your own words, summarize this section in 2–3 sentences:

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8.2 Impulse

CUE / KEY QUESTIONS *Define impulse. What theorem relates it to momentum?* *Why do airbags and padded surfaces reduce injury?*	Impulse (J) is the product of the net force and the time interval it acts over: J = ___________ Impulse–Momentum Theorem: J = ___________ → FΔt = __________________ For a given change in momentum Δp, stretching out the time Δt over which the force acts (e.g., an airbag, a padded floor, bending your knees on landing) ___________________________ needed.

SUMMARY — in your own words, summarize this section in 2–3 sentences:

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8.3 Conservation of Momentum

CUE / KEY QUESTIONS	Law of Conservation of Momentum: the total momentum of a system is __________ (conserved) as long as the net ____________ force on the system is ______.
State the Law of Conservation of Momentum.	In equation form, for a system of two colliding objects: p1i + p2i = ________ + ________
Under what condition does it hold?	
What's the difference between internal and external forces?	Internal forces (between objects inside the system, e.g. during a collision) always cancel in pairs by Newton's Third Law and can never change the ______________ momentum of the system — only an ____________ force can.

SUMMARY — in your own words, summarize this section in 2–3 sentences:

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8.4 Elastic Collisions in One Dimension

CUE / KEY QUESTIONS	An __________ collision is one in which both __________ AND ____________________ are conserved.
Define an elastic collision.	Conservation equations: m1v1i + m2v2i = m1v1f + m2v2f and ½m1v1i² + ½m2v2i² = ____________________
What two quantities are conserved?	
Special case: equal masses, one initially at rest — what happens?	Special case — equal masses, target initially at rest: the incoming object stops completely and the target moves off with the ________________________ (they exchange velocities).

SUMMARY — in your own words, summarize this section in 2–3 sentences:

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8.5 Inelastic Collisions in One Dimension

CUE / KEY QUESTIONS	
Define an inelastic collision.	An __________ collision is one in which momentum is conserved but kinetic energy is ____________ conserved — some KE converts into heat, sound, or deformation.
What is a perfectly (completely) inelastic collision?	A ____________________ collision is the extreme case where the two objects ______________ and move off with one common final velocity — this is when the MOST kinetic energy is lost, for a given set of initial conditions.
Perfectly inelastic collision — final velocity formula?	Perfectly inelastic collision: v′ = ____________________

SUMMARY — in your own words, summarize this section in 2–3 sentences:

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8.6 Collisions of Point Masses in Two Dimensions

CUE / KEY QUESTIONS	
How is momentum conservation applied in 2D?	In two dimensions, momentum is conserved ____________________ — treat the x-components and y-components as two independent 1D conservation problems.
What must be true of the total momentum along each axis?	Σpx,initial = __________ and Σpy,initial = __________
	Typical approach: break every velocity into x- and y-components, apply conservation of momentum along each axis separately, then recombine to find the final speed and __________ (using magnitude and angle).

SUMMARY — in your own words, summarize this section in 2–3 sentences:

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Center of Mass Locating the Center of Mass

CUE / KEY QUESTIONS	
Define center of mass.	The center of mass is the point that moves as though ________________________________ and all external forces acted directly on it.
Equation for the center of mass of a system of point masses (1D)?	For a system of point masses along a line: xcm = __________________________
Why is center-of-mass motion useful for analyzing collisions?	Because internal forces (like collision forces) can't change the system's total momentum, the center of mass of an isolated system moves at ______________________ — even while individual objects inside the system speed up, slow down, or collide.

SUMMARY — in your own words, summarize this section in 2–3 sentences:

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Chapter 6: Uniform Circular Motion — Review

6.1 Rotation Angle and Angular Velocity

CUE / KEY QUESTIONS *Define rotation angle $\Delta\theta$, and its unit.* *Define angular velocity ω.* *Relate linear speed v to angular speed ω.*	Rotation angle: $\Delta\theta$ = ___________, measured in ___________ (2π rad = 360°). Angular velocity: ω = ___________, SI unit: _________________ Relationship between linear and angular speed: v = ______ — points farther from the axis (larger r) move faster in a straight-line sense, for the same ω.

SUMMARY — in your own words, summarize this section in 2–3 sentences:

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6.2 Centripetal Acceleration

CUE / KEY QUESTIONS *Define centripetal acceleration — which direction does it point?* *Give two equivalent equations for ac.*	_____________________ acceleration is the acceleration of an object moving in a circle at constant speed; it always points _____________________ of the circle. ac = ___________ equivalently: ac = ___________ Centripetal acceleration changes the _____________ of the velocity, not its magnitude — that's why uniform circular motion has constant speed but is still accelerating.

SUMMARY — in your own words, summarize this section in 2–3 sentences:

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6.3 Centripetal Force

CUE / KEY QUESTIONS	
Define centripetal force.	____________________ force (Fc) is the net force that causes centripetal acceleration, directed toward the center of the circular path.
Equation for centripetal force (two forms)?	Fc = mac = __________ equivalently: Fc = __________
Is centripetal force a new, separate type of force?	Centripetal force is __________ a new fundamental force — it's just the name for whatever net force (tension, friction, gravity, normal force, etc.) happens to be pointed toward the center and producing the circular motion.

SUMMARY — in your own words, summarize this section in 2–3 sentences:

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Chapter 10: Rotational Motion and Angular Momentum

10.1 Angular Acceleration

CUE / KEY QUESTIONS	
Define angular acceleration α, and its unit. *Relate tangential (linear) acceleration at to α.*	Angular acceleration: α = __________, SI unit: ____________________ Tangential acceleration (the linear acceleration along the direction of motion, tangent to the circle): at = ______ A point on a rotating object can have both a tangential acceleration (speeding up/slowing down) AND a ____________________ acceleration (changing direction) at the same time.

SUMMARY — in your own words, summarize this section in 2–3 sentences:

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10.2 Kinematics of Rotational Motion

CUE / KEY QUESTIONS	
Write the three rotational kinematic equations (constant α). *What does 'rolling without slipping' mean, and what's the key equation?*	Rotational kinematic equations (direct analogs of the linear ones, constant α): ω = ω0 + ______ θ = ω0t + __________ ω² = ω0² + __________ Rolling without slipping: the contact point of a rolling object is momentarily at rest, and the center's linear speed equals v = ______ (and correspondingly a = rα).

SUMMARY — in your own words, summarize this section in 2–3 sentences:

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10.3 Dynamics of Rotational Motion: Rotational Inertia

CUE / KEY QUESTIONS	
Write the rotational analog of Newton's Second Law.	Rotational form of Newton's Second Law: net τ = ______ (torque plays the role of force; moment of inertia plays the role of mass).
Define moment of inertia — what does it depend on?	Moment of inertia (I) measures an object's resistance to a change in rotational motion. It depends on both the object's ________ and how that mass is ____________________ of rotation — mass farther from the axis contributes more.
Point-mass moment of inertia formula?	For a single point mass at radius r from the axis: I = ______ ; for a system of point masses, $I = \Sigma m_i r_i^2$.

SUMMARY — in your own words, summarize this section in 2–3 sentences:

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10.4 Rotational Kinetic Energy: Work and Energy Revisited

CUE / KEY QUESTIONS	
Equation for rotational kinetic energy?	Rotational kinetic energy: KErot = __________
Total kinetic energy of a rolling object?	A rolling object (like a ball or wheel) has both translational and rotational KE at once: KEtotal = __________ + __________
	This is why a rolling object needs MORE total energy to reach the same speed as a sliding (non-rotating) object of the same mass — some of the energy goes into ____________________ as well as moving it forward.

SUMMARY — in your own words, summarize this section in 2–3 sentences:

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10.5 Angular Momentum and Its Conservation

CUE / KEY QUESTIONS	
Define angular momentum L, and its equation.	Angular momentum: L = ____ (the rotational analog of linear momentum p = mv).
State the Law of Conservation of Angular Momentum.	Relation to torque: net τ = ___________ — a net external torque is required to change angular momentum.
Give a real-world example of angular momentum conservation.	Law of Conservation of Angular Momentum: if the net external torque on a system is ______, the system's total angular momentum ______________________.
	Real-world example: ____

SUMMARY — in your own words, summarize this section in 2–3 sentences:

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